1. Use trigonometric substitution to find $\displaystyle\int \frac{1}{x^2\sqrt{x^2-4}}\,dx$.

(Note: You will not receive credit if you use other methods.)

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1-\cos 2x}{2}, \quad \cos^2 x = \frac{1+\cos 2x}{2}, \quad \sin 2x = 2\sin x\cos x$$

$$\int \cot x \csc x\,dx = -\csc x + C, \qquad \int \csc^2 x\,dx = -\cot x + C$$

$$\int \tan x\,dx = \ln|\sec x| + C, \qquad \int \cot x\,dx = \ln|\sin x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C, \qquad \int \csc x\,dx = \ln|\csc x - \cot x| + C$$